Routing and landscape units

Purpose

A routing unit collects the outflow of a group of objects, usually the HRUs of one landscape unit, and routes the sum as a single object. Its row in rout_unit.con sends the result on, typically the total flow to a channel and the recharge to an aquifer. Three files describe routing units: rout_unit.def lists the elements of each unit, rout_unit.ele describes each element, and rout_unit.rtu gives each unit its topography and field records.

A landscape unit is a group of HRUs used only for output. ls_unit.def lists the elements of each landscape unit and ls_unit.ele gives each element its area fractions. The same element file also weights the HRUs in the basin output files. aqu_catunit.ele lists aquifer elements; in 62.0.1 its only effect is to switch on the data rows of the basin aquifer output.

The files are named in file.cio:

File	file.cio line	Slot	Read
rout_unit.def	routing_unit (line 8)	1	when the rtu count in object.cnt is above 0
rout_unit.ele	routing_unit (line 8)	2	when the rtu count is above 0
rout_unit.rtu	routing_unit (line 8)	3	when the rtu count is above 0
ls_unit.ele	regions (line 26)	1	every run
ls_unit.def	regions (line 26)	2	every run
aqu_catunit.ele	regions (line 26)	9	every run, with the calibration inputs

Slot 4 of the routing_unit line (rout_unit.dr) is read from file.cio but never opened.

All six files are free format and whitespace-separated. The readers take the values in column order and do not check the header names.

For rout_unit.rtu, a name of null or a name that does not exist in the run folder means the file is not read. The other five files are skipped when the name is null. When the name is not null and the file does not exist, SWAT+ creates an empty file of that name in the run folder, reads nothing from it, and continues as if the name were null. What a missing file means for the run is given under each file.

rout_unit.def

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line defines one routing unit. The line ends with a list of element numbers, and only the first number of that list has a header.

#	Column	Units	Description
1	id	-	Number of the routing unit. Not used: the rows are taken in order, so row n defines routing unit n.
2	name	-	Name of the routing unit. Not used.
3	elem_tot	-	Number of values in the element list that follows. When the list uses ranges, this is smaller than the number of elements.

#	Column	Units	Description
4 and on	`elements`	-	Element list: `id` values of rows in `rout_unit.ele`, written as described under Element lists.

Element lists

`rout_unit.def` and `ls_unit.def` use the same list format. Each value is an element number, and a negative value closes a range that starts at the value before it. The list `1 -4 7` therefore means elements 1, 2, 3, 4 and 7, and `elem_tot` is 3 because the list has three values. A list with a single value gives one element. An `elem_tot` of 0 gives a unit with no elements.

A negative value must follow a positive one. Two negative values in a row, or a negative first value in a list of more than one value, put negative element numbers into the list.

Behaviour

- Routing unit n is the routing unit whose `id` in `rout_unit.con` is n. It uses row n of `rout_unit.def`, counted from the first data line, whatever the `id` column says. SWAT+ reads as many rows as the `rtu` count in `object.cnt`. Extra rows are ignored, and units without a row have no elements.
- The `define` column of `rout_unit.rtu` is not used to find the row. The routing unit ids in `rout_unit.con`, the rows of `rout_unit.def` and the ids in `rout_unit.rtu` therefore have to run 1, 2, 3 in the same order. The SWAT+ Editor writes them that way.
- An object can be an element of one routing unit only. If two element rows of `rout_unit.ele` point to the same object and both are listed in `rout_unit.def` (in one unit or in two), SWAT+ stops at start-up with an allocation error.
- A routing unit with no elements produces no outflow, and no message is written.
- If `rout_unit.def` is missing or `null` in a project with routing units, every routing unit has no elements. The HRUs then reach the rest of the model only through their rows in `hru.con`, and an HRU without outflow connections there loses its water.

Example

`refdata/Osu_1hru/rout_unit.def` in the SWAT+ repository:

```
rout_unit.def: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id              name  elem_tot  elements
       1             rtu001         1         1
```

Routing unit 1 has one element, element 1 of `rout_unit.ele`.

rout_unit.ele

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one element.

#	Column	Units	Description
1	`id`	-	Element number, used in the element lists of `rout_unit.def`. The rows can be in any order and the numbers do not have to be consecutive; the largest `id` sets the size of the table.
2	`name`	-	Name of the element. Not used.
3	`obj_typ`	-	Object type of the element, one of the codes below.
4	`obj_id`	-	Number of the object within its type: its `id` in the connection file of that type.
5	`frac`	fraction	Fraction of the element in the routing unit, used to scale the element's outflow. See Expansion factor.
6	`dlr`	-	Delivery ratio: `0` or `calc`, `1` or `full`. See Delivery ratio.

The object type codes SWAT+ recognises are:

obj_typ	Object	Connection file
hru	HRU	hru.con
hlt	HRU-lte	hru-lte.con
ru	routing unit	rout_unit.con
cha	legacy channel	channel.con
sdc	SWAT+ channel	chandeg.con
res	reservoir	reservoir.con
exc	export coefficient	exco.con
dr	delivery ratio	delratio.con
out	outlet	outlet.con

Any other code, including **aqu**, **rec** and **rtu**, is not recognised. No message is written, and the element is linked to object 1, the first object in the model (HRU 1 in a model with HRUs). Earlier documentation listed aquifers, recall objects and "another rtu" as possible elements; in 62.0.1 a routing unit inside a routing unit has the code **ru**.

Expansion factor

Each day a routing unit adds up the hydrographs of its elements. Each element's hydrographs are multiplied by an expansion factor e before they are added:

- HRU and HRU-lte elements with **frac** below 0.99999:

$$e = \text{frac}\ \frac{A_{ru}}{A_{obj}}$$

  where A_{ru} is the routing unit's **area** in rout_unit.con and A_{obj} the HRU's **area** in its connection file. When **frac** is the HRU's share of the routing unit area, e is 1 and the whole HRU outflow is added.

- HRU and HRU-lte elements with **frac** of 0.99999 or more: $e = 1$, so the whole HRU outflow is added whatever the areas.
- Export coefficient elements: e as for HRUs when the export coefficient object has an area above 0.01 ha, otherwise 1.
- All other elements: e = frac, the fraction of the object's outflow that enters the routing unit.

The **frac** values are not checked. If they do not add up as intended, the routing unit outflow is scaled accordingly.

The factor and the delivery ratio apply to the total, surface, lateral and tile hydrographs. The recharge hydrograph is multiplied by the factor only. Pesticide, salt and constituent loads are added in full: 62.0.1 applies neither the expansion factor nor the delivery ratio to them.

When another object sends flow to a routing unit through its connection file, the inflow is shared among the unit's elements. Each element receives the sender's fraction multiplied by its **frac**. How an HRU uses that inflow is described under Inflow to HRUs and routing units.

Delivery ratio

The **dlr** column sets a ratio for each hydrograph variable of the element:

dlr	Ratio
0 or calc	Flow 1. All other variables (sediment, nutrients, chlorophyll-a, CBOD, dissolved oxygen, sediment size classes) are multiplied by a calculated ratio, given below.
1 or full	1 for every variable.
any other value	0 for every variable.

For HRU and HRU-lte elements the calculated ratio is

$$DR = \min\left(1,\ \sqrt{\frac{t_{c,\text{elem}}}{t_{c,ru}}}\right)$$

where $t_{c,\text{elem}}$ is the HRU's time of concentration (see Surface runoff lag), or for an HRU-lte the time of concentration of its hru-lte.hru record, and $t_{c,ru}$ is the routing unit's time of concentration (see Time of concentration). For `sdc` and `ru` elements the calculated ratio is 1. For the other element types no ratio is calculated, and the element takes the ratio last calculated for an earlier element (0 if there is none).

A `dlr` value that names a record in `delratio.del` does not work in 62.0.1. The names are looked up when the routing units are set up, and `delratio.del` is read only after that, so no name is ever found. The ratio then stays at 0 for every variable, and the element contributes only its recharge hydrograph. The same happens for any other value, such as `null`. Use `0`, `calc`, `1` or `full`. The SWAT+ Editor writes `0`.

The decision table action `hru_fr_update` (see decision tables) reads a replacement `rout_unit.ele` during the run. By then `delratio.del` has been read, so a `dlr` that names a `delratio.del` record does take that record's ratios from `dr_om.del`. Elements whose `dlr` is `0`, `calc`, `1`, `full` or an unknown name keep the ratios they had before.

Behaviour

- `rout_unit.ele` is required in a project with routing units. If it is missing or `null`, the routing unit tables are never set up and the run fails during start-up.
- In sub-daily runs (time step below one day) with Green and Ampt infiltration (`gampt` = 1 in `codes.bsn`), the routing unit's sub-daily flow is set from each HRU element in turn, so only the last HRU element in the list is kept, and the result is shifted by a fixed 2 hours instead of the routing unit's time of concentration.
- The `ru` row of `print.prt` writes the sum of the element hydrographs of each routing unit to the `ru_*` files (see ru files).

Example

`refdata/0su_1hru/rout_unit.ele`:

```
rout_unit.ele: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                    obj_typ    obj_id          frac               dlr
       1  hru0001                     hru         1             1                 0
```

Element 1 is HRU 1. With `frac` = 1 the whole HRU outflow enters the routing unit. With `dlr` = 0 the water is delivered in full, and sediment and nutrients are multiplied by the calculated delivery ratio.

rout_unit.rtu

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one routing unit.

#	Column	Units	Description
1	`id`	-	Number of the routing unit. Routing unit n (the `id` in `rout_unit.con`) uses the row whose `id` is n. The ids must lie between 1 and the `rtu` count in `object.cnt`. An `id` of 0 ends the file.
2	`name`	-	Name of the routing unit. Not used.
3	`define`	-	Name of the `rout_unit.def` row. Not used: routing unit n always uses row n of `rout_unit.def`.
4	`dlr`	-	Delivery ratio of the routing unit. Not used; delivery ratios are set per element in `rout_unit.ele`.
5	`topo`	-	Name of a record in `topography.hyd`. Its slope and slope length are used for the routing unit's time of concentration.
6	`field`	-	Name of a record in `field.fld`. Its field length is used for the routing unit's time of concentration.

Time of concentration

The routing unit's time of concentration is the sum of an overland flow time and a channel flow time:

$$t_{ov} = \frac{0.0556\,(L_{slp}\,n)^{0.6}}{(slp + 0.001)^{0.3}}, \qquad t_{ch} = \frac{0.62\,L\,n^{0.75}}{A^{0.125}\;(0.5\,(slp + 0.001))^{0.375}}, \qquad t_{c,ru} = t_{ov} + t_{ch}$$

Symbol	Meaning	Units
t_{ov}, t_{ch}, $t_{c,ru}$	overland, channel and total time of concentration	hr
L_{slp}	slope length, `slp_len` of the `topo` record	m
slp	average slope, `slp` of the `topo` record	m/m
L	channel length, `len` of the `field` record divided by 1000	km
A	routing unit area, `area` in `rout_unit.con`	km²
n	Manning's n for overland flow, averaged over the HRU elements	-

The Manning's n is the area-weighted mean of the `ov_mann` values of the HRU elements' land uses (see `landuse.lum`):

$$n = \frac{\sum_i n_i\,A_i}{A}$$

where A_i is the full area of HRU element i; `frac` is not used. When the routing unit has an element that is not an HRU, the running sum is set to 0.1 at that element and the HRU elements after it are added to 0.1. The result then depends on the order of the elements.

The time of concentration is used for the calculated delivery ratio of the elements and, in sub-daily runs, for the unit hydrograph of the routing unit.

Behaviour

- A `topo` name that is `null` or not found in `topography.hyd` gives the first record of that file. A `field` name that is `null` or not found gives the built-in field length of 500 m. No message is written in either case.
- `rout_unit.rtu` is required in a project with routing units. If it is missing or `null`, no routing unit records are set up and the run fails during start-up.

Example

`refdata/Osu_1hru/rout_unit.rtu`:

```
rout_unit.rtu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id              name            define               dlr              topo             field
       1             rtu01             rtu001              null       toportu011            fld011
```

`fld011` is in `field.fld` with a length of 500 m. `toportu011` is not in this dataset's `topography.hyd`, so the routing unit uses its first record, `topohru0001` (slope 0.01078, slope length 10 m), without a message. `define` and `dlr` have no effect.

ls_unit.def

Format

Line 1 is a title, line 2 gives the number of landscape units, and line 3 is the column header. Each later line defines one landscape unit, and the line ends with an element list as in `rout_unit.def`.

#	Column	Units	Description
1	`id`	-	Number of the landscape unit. Not used: row n defines landscape unit n.
2	`name`	-	Name of the landscape unit, written in the `name` column of the landscape unit output files.
3	`area`	ha	Area of the landscape unit. Not used.
4	`elem_tot`	-	Number of values in the element list. 0 gives a landscape unit with no elements, whose output is all zeros.
5 and on	`elements`	-	Element list: `id` values of rows in `ls_unit.ele`, written as described under Element lists.

SWAT+ reads as many rows as line 2 gives.

Behaviour

- Landscape units are used for output only. Each one gets a row in the `lsunit_wb`, `lsunit_nb`, `lsunit_ls` and `lsunit_pw` files and in the landscape unit carbon files (`lsu_cb_gl`, `lsu_cb_trf`, `lsu_cb_plt`), switched on in `print.prt`. Each value is the sum over the unit's elements of the HRU (or HRU-lte) value multiplied by the element's `sub_frac` in `ls_unit.ele`. See Aggregation.
- The element number is used as the HRU number: element n adds the output of HRU n (or HRU-lte n), and its `obj_typ_no` is not consulted. The element ids in `ls_unit.ele` therefore have to equal the HRU numbers.
- In the output files, `unit` is the landscape unit number and `gis_id` is taken from the routing unit with the same number in `rout_unit.con`.
- With gwflow and landscape unit connection (`conn_type` 2, see `lsucell.gw`), the element lists also decide which HRUs pass recharge and remaining evapotranspiration to the cells of each landscape unit. Here too element n is HRU n.
- `ls_unit.def` needs `ls_unit.ele`. If `ls_unit.def` is missing or `null`, the project has no landscape units and the landscape unit output files are not opened.

Example

`refdata/Osu_1hru/ls_unit.def`:

```
ls_unit.def: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
1
      id              name          area  elem_tot  elements
       1            rtu001       10.0000         1         1
```

One landscape unit of 10 ha, named `rtu001` after the routing unit, with element 1 of `ls_unit.ele`.

ls_unit.ele

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one element. Six values are read from each line.

#	Column	Units	Description
1	`id`	-	Element number, used in the element lists of `ls_unit.def`. The largest `id` sets the size of the table. Must equal the HRU number (see below).
2	`name`	-	Name of the element. Not used.
3	`obj_typ`	-	`hru` or `hlt`. Elements of any other type are skipped in the sums.
4	`obj_typ_no`	-	HRU or HRU-lte number.
5	`bsn_frac`	fraction	Fraction of the basin land area (`ls_area` in `object.cnt`) covered by the HRU. Weight of the HRU in the basin output files.

#	Column	Units	Description
6	`sub_frac`	fraction	Fraction of the landscape unit area covered by the HRU. Weight of the HRU in the landscape unit output files. Called `lsu_frac` in earlier documentation; the `Ames_sub1` file labels it `RU_FRAC`.

Behaviour

- Basin output: the `basin_wb`, `basin_nb`, `basin_ls` and `basin_pw` files, including the starting soil water and snow in `basin_wb`, are the sums of the HRU values multiplied by `bsn_frac`. The fractions are used as given; if they do not add up to 1, the basin values are scaled by their sum.
- For each HRU number n, the basin sum reads row n, takes its `obj_typ_no` m, and then uses the `obj_typ` and `bsn_frac` of row m with the output of HRU m. The sum is right when row n describes HRU n, which is how the SWAT+ Editor writes the file. HRU-lte objects are summed the same way over the first rows of the file.
- Landscape unit output uses `sub_frac` (see `ls_unit.def`).
- The soft calibration of the whole basin in `water_balance.sft` takes each HRU's area as `ls_area` times its `bsn_frac` (see soft calibration).
- The decision table action `hru_fr_update` reads a replacement `ls_unit.ele` during the run and sets the area of each `hru` element to `bsn_frac` times `ls_area`.
- If `ls_unit.ele` is missing or `null`, the basin landscape files are opened and get their headers, and no data rows are written. The routine that stores the basin's starting soil water at the end of the warm-up years still reads the element table, so a model with HRUs and no `ls_unit.ele` can stop with a memory error at that point. Treat the file as required in any model with HRUs.

Example

`refdata/Ames_sub1/ls_unit.ele`:

```
lsu_unit.ele
           ID    NAME      OBJ_TYP  OBJ_TYP_NO    BSN_FRAC       RU_FRAC
            1    hru0001      hru           1     0.20           0.0
            2    hru0002      hru           2     0.20           0.0
            3    hru0003      hru           3     0.20           0.0
            4    hru0004      hru           4     0.20           0.0
            5    hru0005      hru           5     0.10           0.0
            6    hru0006      hru           6     0.10           0.0
            7    hru0007      hru           7     0.20           0.0
            8    hru0008      hru           8     0.20           0.0
            9    hru0009      hru           9     0.20           0.0
           10    hru0010      hru          10     0.20           0.0
           11    hru0011      hru          11     0.10           0.0
           12    hru0012      hru          12     0.10           0.0
```

This file is not in the SWAT+ Editor layout: it has six columns and upper-case headers. Each row describes the HRU of the same number, as the basin sums require. The `BSN_FRAC` values add up to 2.0, so every basin output of this dataset is twice the area-weighted HRU average. `Ames_sub1` has no `ls_unit.def`, so the `RU_FRAC` values of 0 are not used.

aqu_catunit.ele

Format

Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one element, and seven values are read from it. A line with fewer values takes the missing ones from the next line.

#	Column	Units	Description
1	id	-	Element number. The largest id sets the size of the table.
2	name	-	Name of the element.
3	obj_typ	-	Object type, aqu in files written by the SWAT+ Editor.
4	obj_typ_no	-	Aquifer number.
5	bsn_frac	fraction	Fraction of the basin covered by the aquifer.
6	sub_frac	fraction	Fraction of the aquifer region covered by the element.
7	reg_frac	fraction	Fraction of the calibration region covered by the element.

Behaviour

- In 62.0.1 the file works as a switch. When it has at least one element, the basin aquifer files (basin_aqu row in print.prt) get one data row per interval. Without it they get headers only.
- The basin aquifer values are weighted by each aquifer's area divided by tot_area in object.cnt (see aquifer output). The columns of aqu_catunit.ele are not used for this.
- The columns are meant for aquifer regions defined in aqu_catunit.def and aqu_reg.def. SWAT+ 62.0.1 reads those files, but no calculation or output uses the regions.

Example

The first rows of refdata/Osu_1hru/aqu_catunit.ele:

```
aqu_catunit.ele: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                    obj_typ  obj_typ_no      bsn_frac      sub_frac      reg_frac
       1  aqu011                      aqu           1       0.00241       0.00000       0.00000
       2  aqu012                      aqu           2       0.10460       0.00000       0.00000
       3  aqu021                      aqu           3       0.01290       0.00000       0.00000
```

The file has 21 elements, while the dataset has one aquifer. The rows have no effect beyond switching on the basin aquifer output. In refdata/Ames_sub1 the file is empty, so that dataset writes no basin aquifer data rows.

Columns in earlier documentation

- ls_unit.ele: the SWAT+ Editor writes a seventh column, reg_frac. SWAT+ 62.0.1 reads six values per line and ignores it.
- ls_unit.def: earlier documentation did not list the line with the number of landscape units (line 2) or the area column. Both are read, and a file without them is read out of step.

Related files

- rout_unit.con: the routing unit objects, their areas and their outflow connections.
- hru.con and hru-data.hru: the HRUs that are elements of routing and landscape units.
- object.cnt: the rtu count, and ls_area and tot_area used with bsn_frac and the basin aquifer output.
- file.cio: names the files on the routing_unit and regions lines.
- topography.hyd and field.fld: the topo and field records of rout_unit.rtu.
- aquifer.aqu: the aquifers listed in aqu_catunit.ele.
- print.prt: switches for the ru, lsunit_*, basin_* and basin_aqu outputs.
- Decision tables: the hru_fr_update action.
- Outputs: water balance, nutrient balance and losses, hydrographs, aquifer output.
- Delivery ratios in the model theory.

Sources

- File name slots: type input_ru and type input_regions in input_file_module.f90, read in readcio_read.f90; read order in main.f90.in and hyd_connect.f90
- Routing unit readers: ru_read.f90 (rout_unit.rtu), ru_read_elements.f90 (rout_unit.ele, rout_unit.def); element lists: define_unit_elements.f90

- Routing unit types: `ru_parameters` in ru_module.f90, `routing_unit_data` and `routing_unit_elements` in hydrograph_module.f90
- Routing unit linkage and inflow sharing: hyd_connect.f90, hyd_read_connect.f90
- Daily routing unit sum: ru_control.f90, called from command.f90
- Delivery ratios: dr_ru.f90, dr_db_read.f90
- Time of concentration and unit hydrographs: time_conc_init.f90, unit_hyd_ru_hru.f90
- Landscape unit reader: lsu_read_elements.f90; types `landscape_units` and `landscape_elements` in calibration_data_module.f90
- Landscape unit and basin output: lsu_output.f90, lsu_carbon_output.f90, basin_output.f90, basin_sw_init.f90 (called from time_control.f90), file opening in output_landscape_init.f90
- Other uses of `ls_unit.ele` and `ls_unit.def`: lcu_read_softcal.f90, hru_fr_change.f90 (action `hru_fr_update` in actions.f90), gwflow_rech.f90, gwflow_gwet.f90
- Aquifer elements: aqu_read_elements.f90, called from proc_cal.f90; basin aquifer output in basin_aquifer_output.f90, files opened in header_write.f90

File	Column	Fortran variable
rout_unit.def	id	not stored
rout_unit.def	name	ru_def%name
rout_unit.def	elem_tot	nspu
rout_unit.def	elements	elem_cnt, expanded into ru_def%num and ru_def%num_tot
rout_unit.ele	id	index of ru_elem
rout_unit.ele	name	ru_elem%name
rout_unit.ele	obj_typ	ru_elem%obtyp
rout_unit.ele	obj_id	ru_elem%obtypno (object number in ru_elem%obj)
rout_unit.ele	frac	ru_elem%frac
rout_unit.ele	dlr	ru_elem%dr_name (ratios in ru_elem%dr)
rout_unit.rtu	id	index of ru
rout_unit.rtu	name	ru%name
rout_unit.rtu	define	ru%dbsc%elem_def
rout_unit.rtu	dlr	ru%dbsc%elem_dr
rout_unit.rtu	topo	ru%dbsc%toposub_db (index ru%dbs%toposub_db)
rout_unit.rtu	field	ru%dbsc%field_db (index ru%dbs%field_db, values in ru%field)
ls_unit.def	line 2	mlsu, stored as db_mx%lsu_out
ls_unit.def	name	lsu_out%name
ls_unit.def	area	lsu_out%area_ha
ls_unit.def	elements	lsu_out%num, lsu_out%num_tot
ls_unit.ele	obj_typ	lsu_elem%obtyp
ls_unit.ele	obj_typ_no	lsu_elem%obtypno
ls_unit.ele	bsn_frac	lsu_elem%bsn_frac
ls_unit.ele	sub_frac	lsu_elem%ru_frac
aqu_catunit.ele	obj_typ, obj_typ_no	acu_elem%obtyp, acu_elem%obtypno
aqu_catunit.ele	bsn_frac, sub_frac, reg_frac	acu_elem%bsn_frac, acu_elem%ru_frac, acu_elem%reg_frac
aqu_catunit.ele	largest id	db_mx%aqu_elem